
The Impact of Urban Road Closures on Public Transit

Case Study: Jerusalem

Course 71253 — Introduction to Data Science
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Aza Street closures make Rehavia's buses untrustable

Aza is a narrow, central traffic artery

The Prime Minister lives on Aza Street.

Therefore ,Rehavia is a hub of ongoing protests.

Police repeatedly close the street to traffic.

We study lines that pass through this area, affected by closures

מקטע מלא ברחוב עזה	17
	19
	22
מקטע חלקי ברחוב עזה	9
	97
קווי ביקורת	14
	15

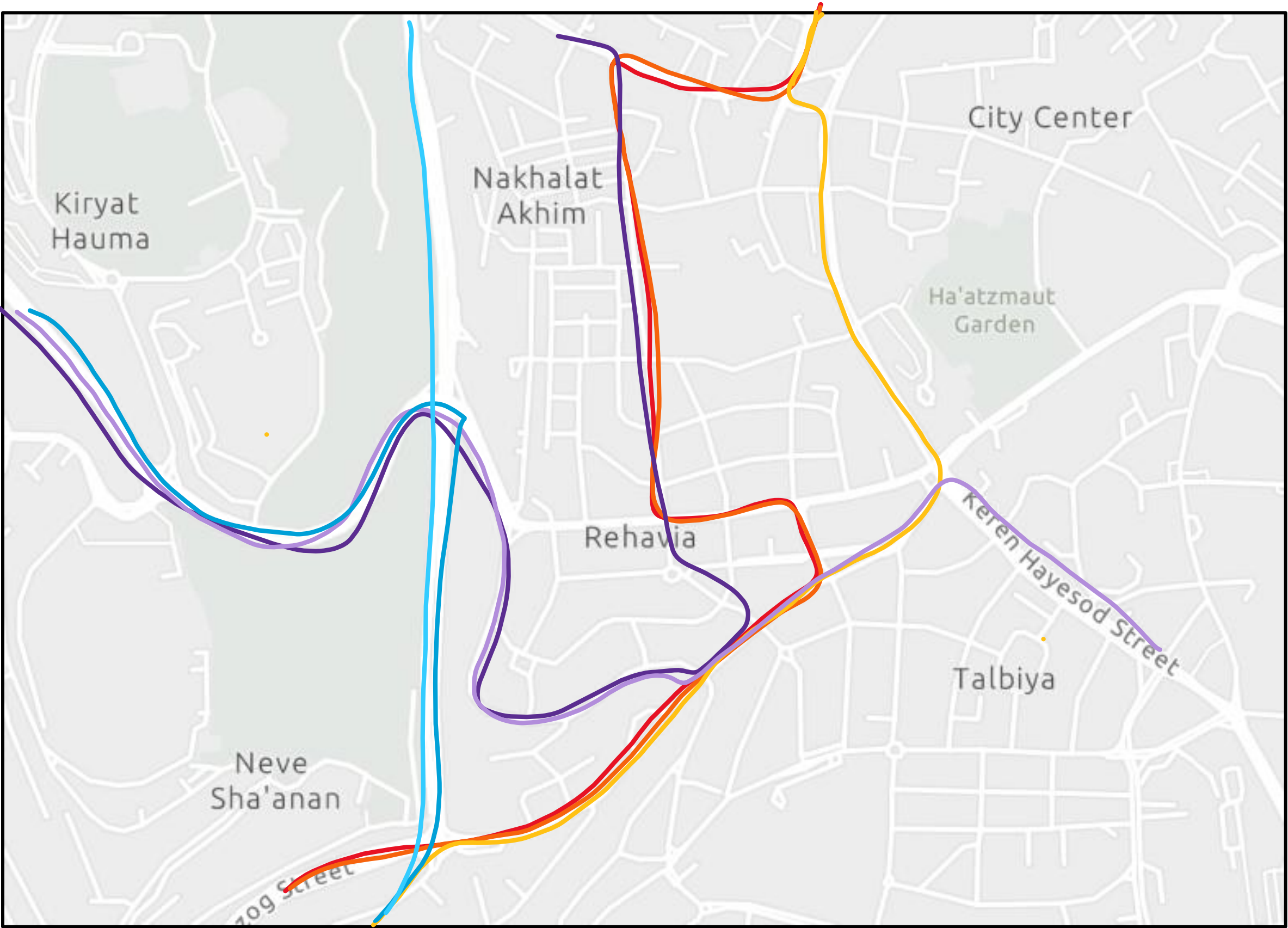


FIGURE 1 — MAP OF REHAVIA AND THE STUDIED BUS LINES

Exploratory analysis

Ministry of Transport dataset, 2024. 7 bus lines.
Each row: one line, month, day of the week , hour, stop.
A full trip ("block") is one sequence of rows.

cumulative_distance (m)	cumulative_time (min)	n_observations	Stop name	stop_sequence
0	1.46	10	דולצ'ין/קוליץ	1
1,926	4.96	10	הכפר השוודי/סולד	2
2,244	6.28	10	מרכז פיליפ ליאון/סולד	3

TABLE 1 — EXAMPLE BLOCK, LINE 19 (SUNDAY, 5AM DEPARTURE)

Cleaning cut duplicates, not real data

708,642 → 603,662 records after removing duplicates

6.3% of rows flagged low-confidence, not deleted

25/26 → 1/2 hours remapped to midnight–2am

A minority of trips skip stops on Aza St.

Each line has a reference route — its most common path.

A minority of trips skip a segment of stops.

These are the line's route "variants."

cumulative_distance (m)	cumulative_time (min)	n_observations	stop_name	stop_sequence
8,249	21.11	7	Herzog/Tchernichovsky	16
8,453	25.94	7	Ussishkin/Bezalel	23

TableE 2 — SKIPPED STOPS ON THE AZA ST. DETOUR-example data

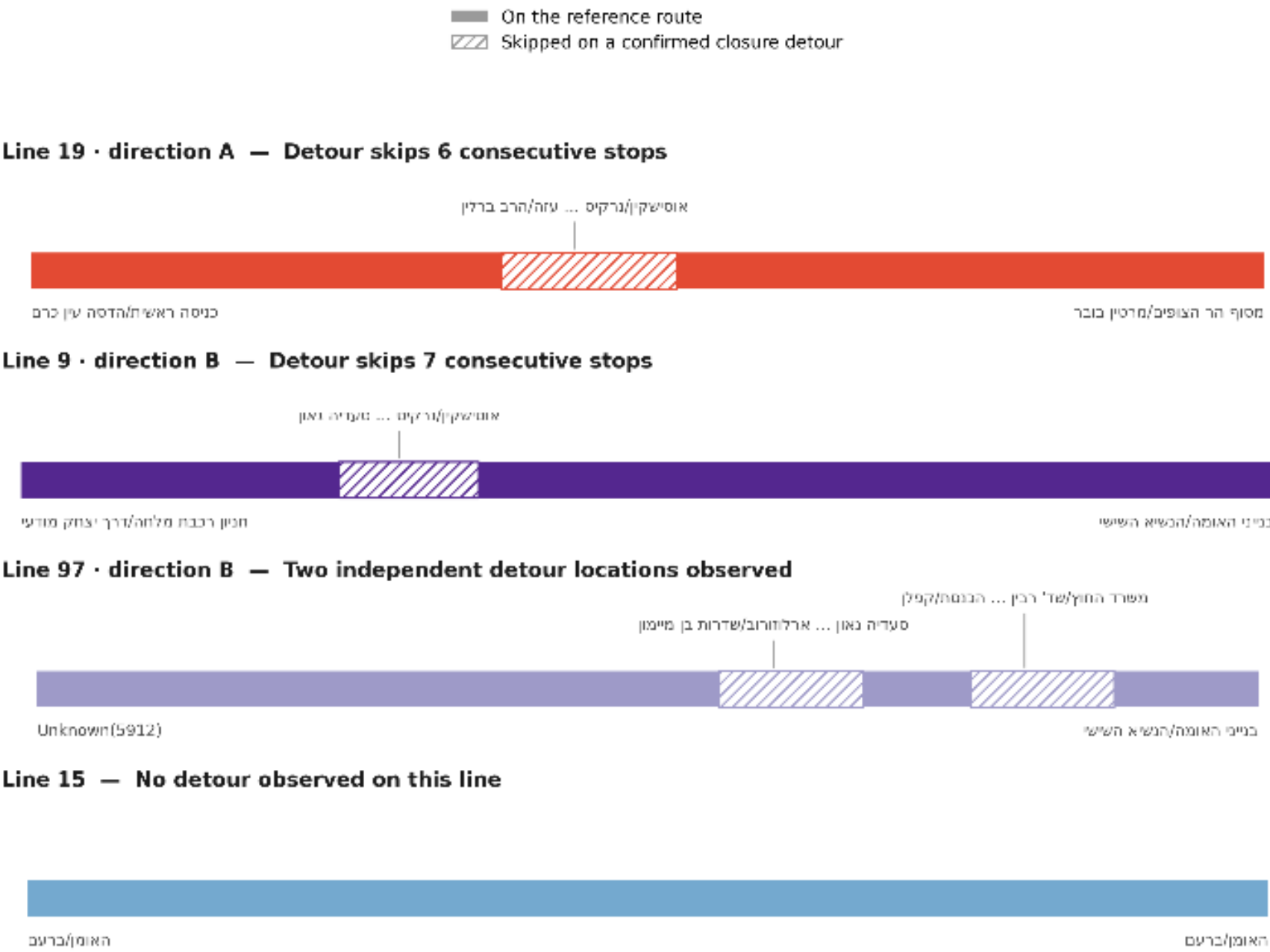


FIGURE 2 — SKIPPED STOPS ON THE AZA ST. DETOUR

Line 22B always completes the full route —excluded from this analysis.

How can a resident know the most reliable way out?

- 1 When and how often do blockages occur on Aza Street?
- 2 Does the detour make the trip longer or shorter?
- 3 Do blockages also affect lines that skip Aza Street?

travel time of reference routes

Longer routes take longer: Line 9 ~20 km/90 min, Line 97 ~10 km/44 min.
All seven lines share one daily rhythm — rising, peaking at midday, dropping by evening.

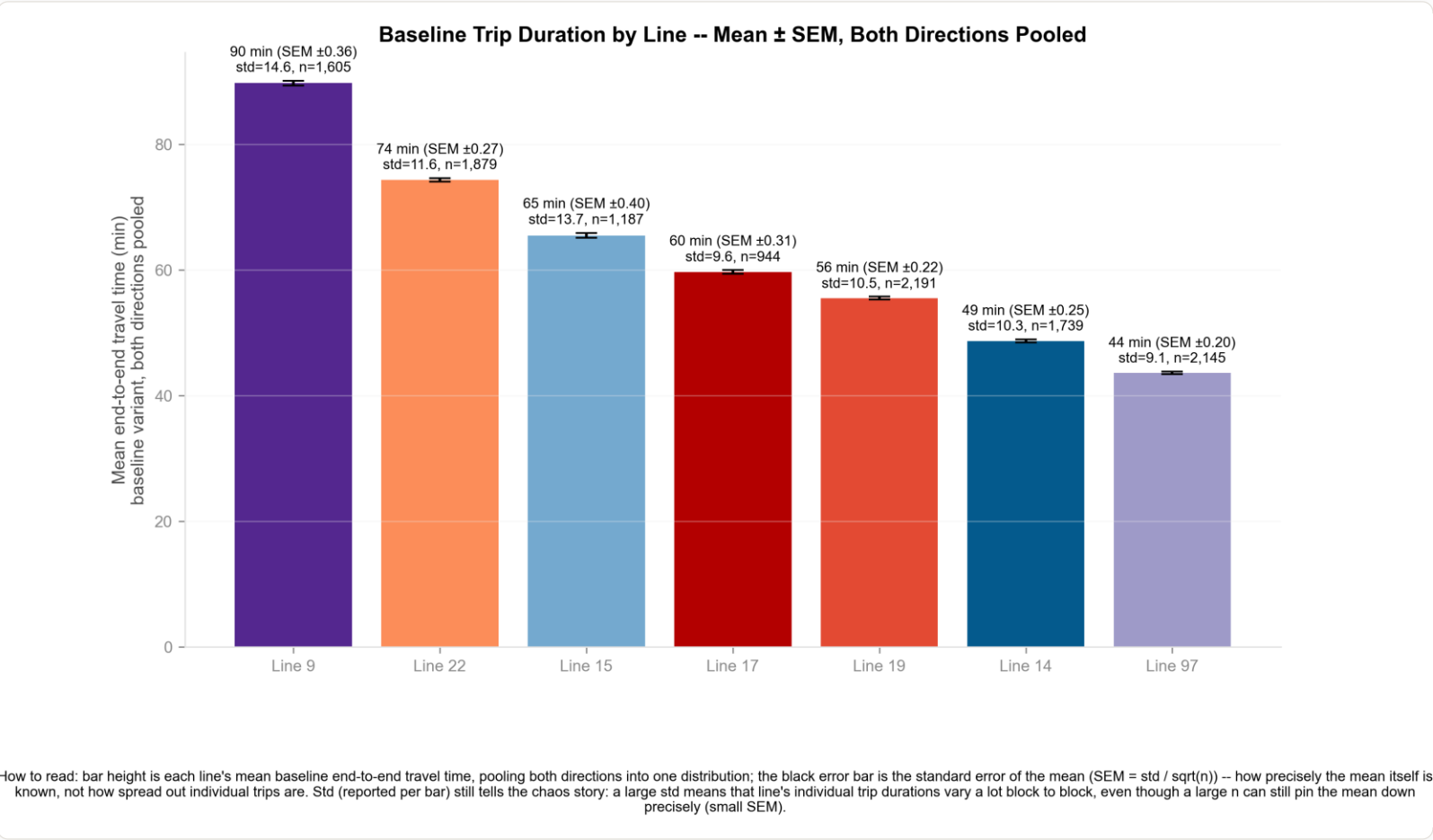


FIGURE 3A — BASELINE TRIP DURATION BY LINE

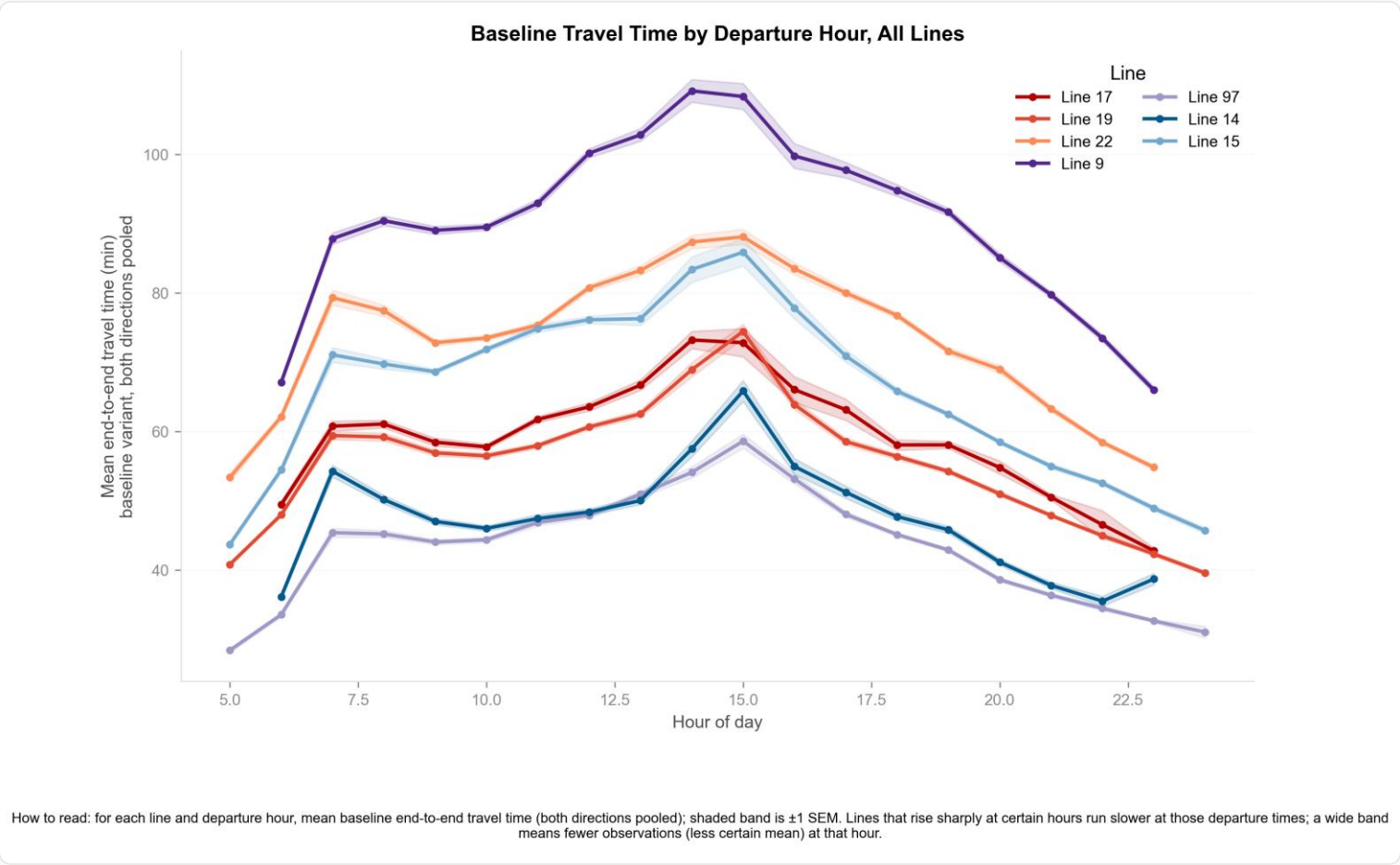


FIGURE 3B — BASELINE TRAVEL TIME BY DEPARTURE HOUR

When does Blockages occur

5%–12% of corridor trips detour on an average weekday.- at least 3 lines run on blockadge variants , same time
Saturday nights jump to 66%–84%.
Control lines 14/15 stay near 0%.

21 confirmed blockage events, mostly Saturdays

11 of 21 events are Saturday nights, 7pm–midnight.
The rest cluster on weekdays.

On confirmed events, the data indicates
that the duration of the blockages differ
between lines

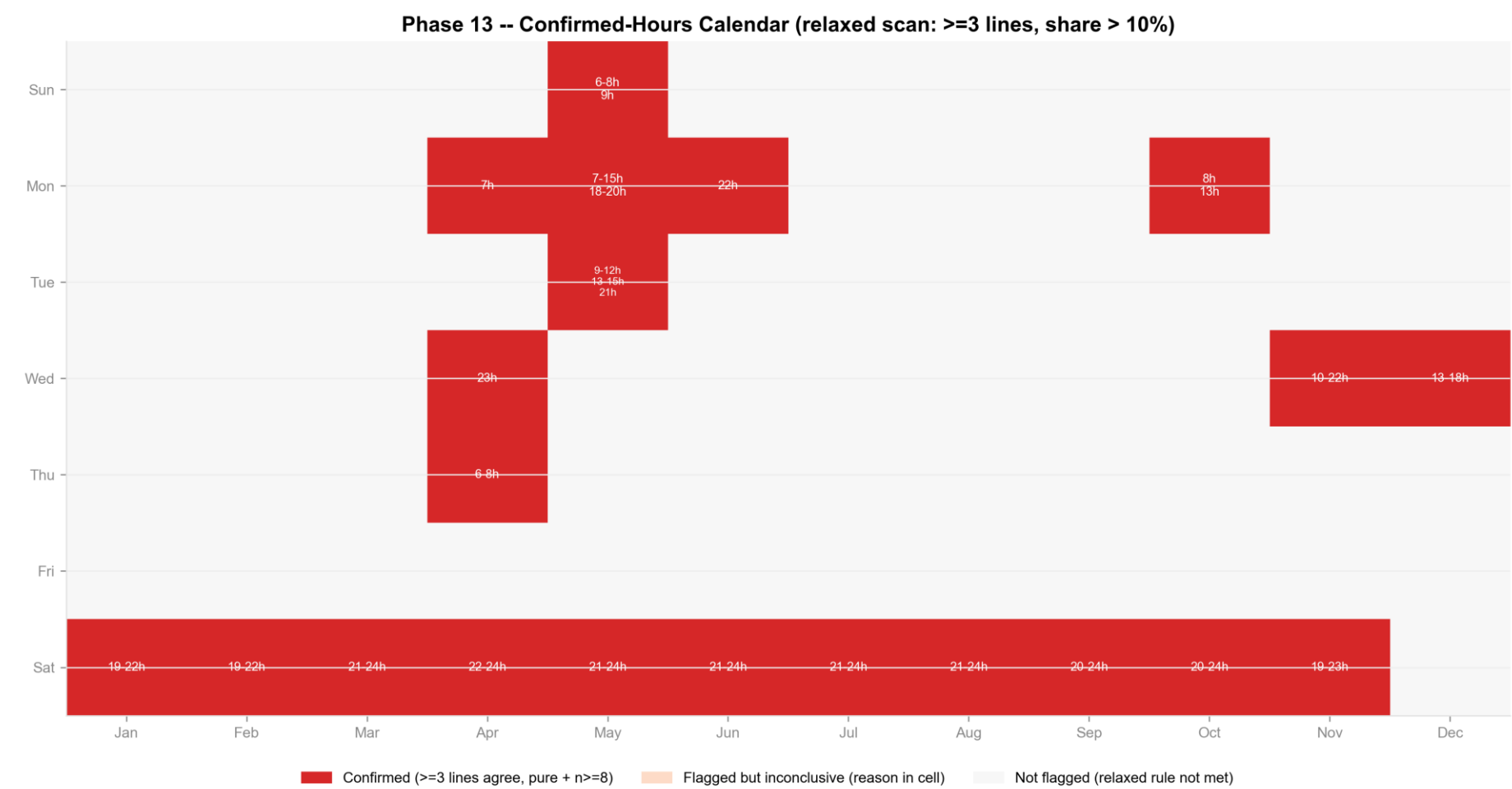


FIGURE 6 — CONFIRMED BLOCKAGE HOURS BY MONTH AND DAY

Treval time of The detour varient

On average, detours save 8 to 16 minutes.

Statistically significant for all 5 detouring lines.

During rush hour, the advantage disappears.

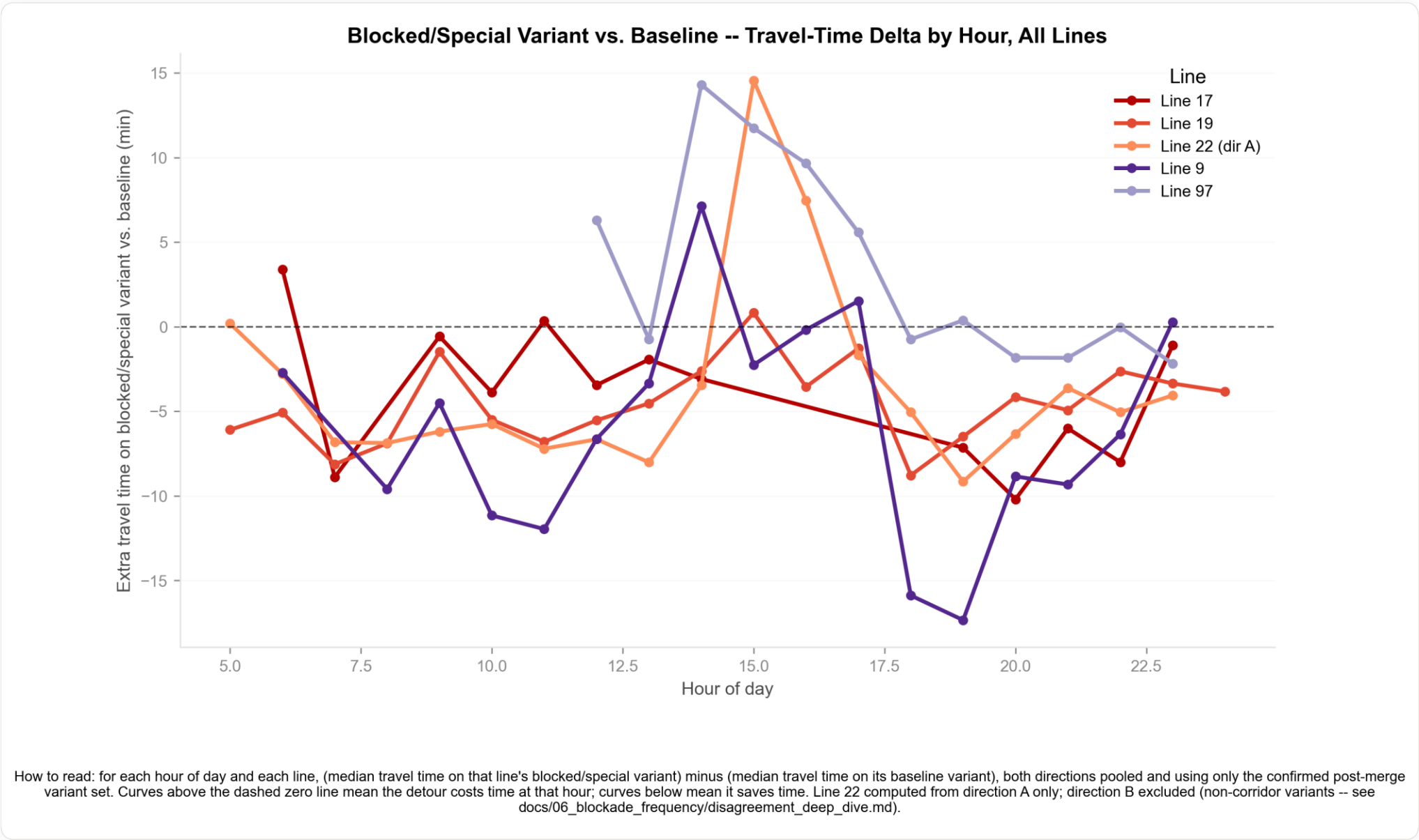


FIGURE 7 — MEDIAN TRAVEL-TIME DELTA BY HOUR

Effect of Blockages on the control lines 14/15

Bolcadge – only 1 line is using the blocked variant

Mann-Whitney and permutation tests: no significant difference.

Their travel times are unaffected by the closures.

p t Welch's	p Mann-Whitney	Δ (min)	Median normal (min)	Median blockage (min)	n normal	n blockage	Stratum	Line
0.353	0.376	0.9	52	52.9	12	33	Saturday	15
0.543	0.941	1.8–	66.6	64.8	746	153	Weekday	15
0.256	0.317	1.2	36.6	37.8	7	40	Saturday	14
0.064	0.156	0.9	47.1	48	1,108	198	Weekday	14

TABLE 4 – CONTROL LINES' TRAVEL TIME, BLOCKADE VS. NORMAL

Summery

לא מצאנו הבדלים לרעה על התחבורה הציבורית בירושלים בזמן חסימות אפשר לומר שחסימת רחוב ראשי לא בהכרח גורמת לעומס מיוחד על הכבישין נסביב

הנתונים מעידים על כאוס , אין סדר , אין לדעת מה המסלול בו השתמשו האוטובסים. כאשר יש חסימה אמנם האוטובוס מהיר יותר אבל האזרח צריך ללכת יותר , וכדי להשלים מידע על זה צריך להשיג מידע על משתמשי אוטובוס שהוא אינו זמין . או לא מצאנו אותו

ותכלס זה בעיקר מה שחסר לנו כדי להשלים את התמונה